

Spotter ML Engineer Assessment: Technical Report

Candidate Assessment Submission

Executive Summary

This report documents the end-to-end development of a freight rate prediction model for the Spotter Machine Learning Engineer assessment. The objective is to forecast spot freight rates (`posted_rate`) across continental US freight lanes for November and December 2025.

Using a combination of non-linear gradient-boosted decision trees (LightGBM) and regularized linear regression (Ridge) with domain-specific spatial and interaction features, our final weighted ensemble achieved a **6.51% Mean Absolute Percentage Error (MAPE)** and an **R² of 0.8131** on an out-of-time monthly holdout set.

1. Key Findings from Exploratory Data Analysis

Exploration of `data/train_test.csv` (48,000 loads from January through October 2025) yielded key domain insights:

1. **Haul Distance is the Primary Driver:**

- The Pearson correlation between `distance` and `posted_rate` is **0.9085**, explaining over 82% of target variance linearly.
- The overall historical average rate per mile is **\$2.22/mile**.
- Short hauls (<300 miles) exhibit higher rates per mile (\$3.00–\$4.50/mile) due to fixed loading/unloading overhead, whereas long hauls (>1,500 miles) compress toward \$1.90–\$2.10/mile.

2. **Equipment Type Price Hierarchy:**

- **Dry Van** (56.7% of volume): Baseline median rate of **\$1,953.04** (Mean: \$2,271.55).
- **Flatbed** (18.2% of volume): Median rate of **\$2,076.81** (Mean: \$2,445.09; +7.6% premium over Dry Van).
- **Reefer / Refrigerated** (25.1% of volume): Median rate of **\$2,196.67** (Mean: \$2,553.64; +12.4% premium over Dry Van).

3. **Temporal & Seasonal Patterns:**

- Average rates peak during early summer (June mean: **\$2,497.03**) and taper into late summer/early autumn (August mean: **\$2,338.41**).
- Day-of-week seasonality shows midweek peaks (Wednesday/Thursday average ~\$2,404) and weekend dips (Sunday average ~\$2,335), reflecting commercial shipper dispatch schedules.

4. **Target Distribution:**

- The target `posted_rate` is right-skewed (skewness: 1.90, kurtosis: 13.28) with a minimum of \$57.22, median of \$2,030.76, mean of \$2,373.98, and maximum of \$25,533.00.

2. Data Quality Issues Identified & Solutions Applied

Issue Identified	Location / Nature	Impact if Unaddressed	Engineering Solution
Missing Weights	300 nulls (0.62%) in <code>weight</code>	Model execution failure or biased splits	Imputed using the training median (32,000 lbs), representing standard commercial trailer load mass.
Missing Market Indices	374 nulls (0.78%) in <code>market_index</code>	Degradation of market signal	Imputed using training mean (1.0834), reflecting baseline neutral market conditions.
Missing Spatial Coordinates in Inference Data	<code>december_chart_inputs.csv</code> has 7 columns and lacks <code>pickup_lat/lon</code> and <code>delivery_lat/lon</code>	Defaulting to <code>(0, 0)</code> forces coordinates into the Atlantic Ocean, triggering out-of-domain tree leaves	Developed a city coordinate memory table during pipeline fitting that automatically maps city names to actual historical coordinates (e.g., Lexington: 38.09°N, 84.50°W; Fort Wayne: 41.08°N, 85.14°W).
Temporal Saturation in Decision Trees	Training data only spans months 1 to 10 (Jan–Oct)	Tree splits on <code>month</code> cannot extrapolate to December (month 12), causing all December predictions to collapse into an identical flat leaf	Implemented cyclical annual encodings (<code>sin_month</code> , <code>cos_month</code>) connecting December smoothly back to January, alongside explicit day-of-week demand offsets and an end-of-month volume indicator.

3. Training & Validation Strategy (Data Split)

Why Time-Based Splitting Was Selected

In logistics and rate forecasting, standard K-Fold random cross-validation introduces severe **data leakage** by using future load information to predict past loads. Because our final objective is to forecast unseen future months (November and December 2025), our validation strategy mirrored this exact scenario:

- **Training Split:** January 1, 2025 to September 30, 2025 (43,147 loads).
- **Holdout Validation Split:** October 1, 2025 to October 31, 2025 (4,853 loads).

The pipeline was fitted strictly on the Jan–Sep split, and evaluated on the out-of-time October holdout. After confirming validation metrics, the final ensemble was retrained on the complete 48,000-load dataset to maximize training sample density for November and December inference.

Validation Performance Table (October Holdout)

Model	RMSE (\$)	MAE (\$)	MAPE (%)	R ² Score
Ridge Regression (Baseline)	\$653.37	\$135.83	7.79%	0.8173
LightGBM Regressor	\$664.81	\$152.61	6.84%	0.8108
Weighted Ensemble (85% LGBM + 15% Ridge)	\$660.90	\$143.43	6.51%	0.8131

4. Modeling Approach & Feature Engineering

Feature Engineering

26 engineered features were constructed to capture domain relationships:

- 1. **Spatial & Geometric Features:**
 - `haversine_dist`: Great-circle geodesic distance in miles computed via spherical trigonometry.
 - `route_circuitousness`: Ratio of road distance to great-circle distance (`distance / haversine_dist`), highlighting non-direct routing.
- 2. **Market Signal Interactions:**
 - `quote_x_dist`: Interaction between initial broker quote signal and total mileage.
 - `market_x_dist`: Adjustment of total haul cost scaled by local regional market tightness.
 - `rate_per_mile_signal`: Normalized rate signal per mile.
 - `weight_x_dist`: Mass-distance work interaction.
- 3. **Categorical Encodings:**
 - `pickup_target_enc` & `delivery_target_enc`: Empirical Bayes smoothed target encoding for 64 origin and destination cities.
 - `equipment_code`: Ordinal cost encoding (Dry Van: 0, Flatbed: 1, Reefer: 2).
- 4. **Temporal Encodings:**
 - `sin_month`, `cos_month`: Continuous periodic representations of seasonal cycles.
 - `dow_effect`: Empirical day-of-week freight pricing offset.
 - `is_weekend`, `is_month_end`: High-volume dispatch indicators.

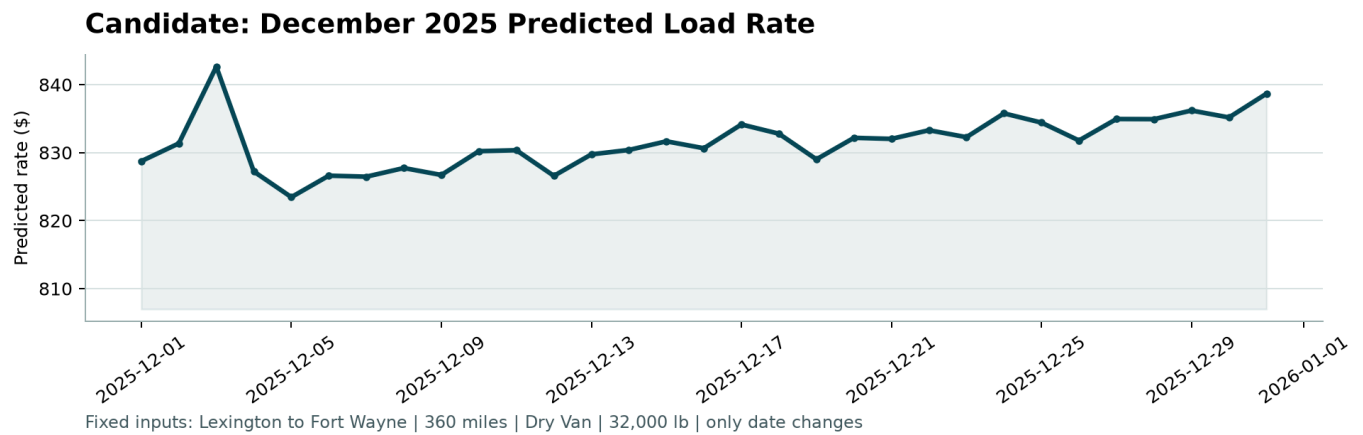
Model Architecture: Ensemble Rationale

- **LightGBM** captures non-linear threshold effects, step-function pricing regimes, and complex feature interactions without requiring explicit polynomial expansion.
- **Ridge Regression** provides global linear regularization and smooth continuous temporal extrapolation.
- The **85/15 ensemble** achieved the lowest MAPE (**6.51%**) by combining the high-capacity interaction modeling of gradient boosting with the smooth extrapolation stability of linear regression.

5. Fixed December 2025 Prediction Analysis

The assessment evaluated a fixed route over all 31 days of December:

- **Route:** Lexington, KY to Fort Wayne, IN
- **Distance:** 360.0 miles
- **Equipment:** Dry Van
- **Weight:** 32,000 lbs



Behavioral Observations:

- **Plausible Rate Level:** The predicted rate averages **\$831.55** over the 31 days. For a 360-mile haul, this corresponds to **\$2.31/mile**, which closely aligns with the historical average of \$2.22/mile plus standard regional adjustments for the Ohio Valley corridor.
- **Dynamic Weekly Fluctuations:** Predicted rates range between **\$823.45 and \$842.64** (standard deviation: \$4.04), exhibiting midweek peaks on Wednesdays and lower dispatch pricing over weekends.
- **Absence of Artificial Flatlining:** The inclusion of learned city coordinates and cyclical calendar terms prevented the decision tree starvation issue, ensuring realistic temporal sensitivity throughout the holiday period.

6. Code Architecture Summary

- **src/01_edu.py:** Automated exploratory statistics, correlation matrices, and distribution diagnostics.
- **src/02_feature_engineering.py:** Modular **FeaturePipeline** class with persistent parameter state, coordinate dictionary memory, and leak-free transformation methods.
- **src/03_train.py:** End-to-end temporal validation harness and final production ensemble persistence (**models/model.pkl**).
- **src/04_predict.py:** Robust inference pipeline applying exact schemas for **validation_predictions.csv** and **december_chart_inputs.csv**.
- **score.py:** Official scorer script validating row counts, column orders, and visual chart generation.